Machine Learning

Support Vector Machine, Part 2

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Contents

- **What is SVM ?**
- **Objective function of SVM**
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Primal Problem

- **Step 1: Solve the minimization problem** (primal form)

$$\min_{\mathbf{w}, b} L_P(\mathbf{w}, b, \boldsymbol{\alpha})$$

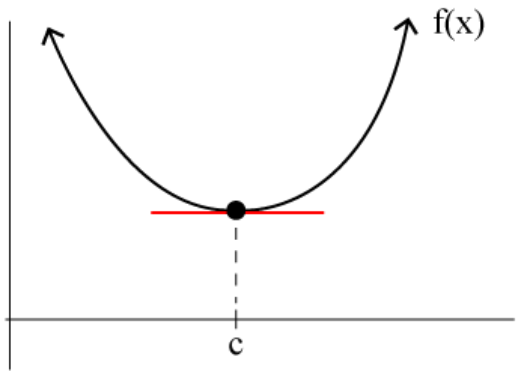
where $L_P(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{||\mathbf{w}||^2}{2} - \sum_{i=1}^{N} \alpha_i \cdot \{y_i(\mathbf{w}^{\mathbf{T}}\mathbf{x}_i + b) - 1\}$

- The form is convex & continuous with respect to $\mathbf{w}, b$
- So, we can directly obtain the optimal $\mathbf{w}^*$ and b^* through:

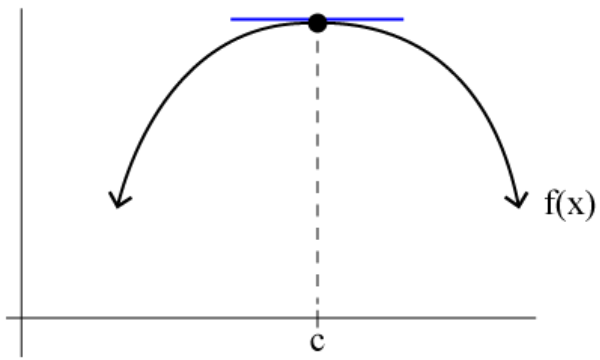
$$\frac{\partial L_p}{\partial \mathbf{w}} = 0 \quad \Rightarrow \quad \mathbf{w}^* = \sum_{i=1}^{N} \alpha_i \, y_i \mathbf{x}_i \qquad \frac{\partial L_p}{\partial b} = 0 \quad \Rightarrow \quad \sum_{i=1}^{N} \alpha_i y_i = 0$$

(Appendix) Hessian Matrix

- **How can we determine whether a function f is convex or not?**
  - Single variable function: if the second derivatives of the function is positive => $f''(x) > 0$, it is convex.



$f'(c) = 0$ **and** $f''(c) > 0$



$f'(c) = 0$ **and** $f''(c) < 0$

  - Multivariate function: determining based on whether its **Hessian matrix** is **positive semi-definite (PSD)**.

$$f : R^n \rightarrow R$$

$$H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & & & \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

(Appendix) Hessian Matrix

- **Positive semi-definite (PSD) of a matrix H**
  - If $\mathbf{x}^{\mathbf{T}} H \mathbf{x} \geq 0$ for an arbitrary $\mathbf{x}$, H is positive semi-definite.
- **Exercise:**
  - Determine whether the function $f(\mathbf{w}) = \frac{||\mathbf{w}||^2}{2}$ is convex based on its Hessian matrix (here, $\mathbf{w} \in \boldsymbol{R}^{\boldsymbol{n}}$)
  - Solution:
    - Its Hessian matrix is the identity matrix, I_n
    - Let $\mathbf{x}^{\mathbf{T}} = [x_1, x_2, \dots, x_n]$.

$$\mathbf{x}^{\mathbf{T}} H \mathbf{x} = \mathbf{x}^{\mathbf{T}} I_n \mathbf{x} = \mathbf{x}^{\mathbf{T}} \mathbf{x} = ||\mathbf{x}||^2$$

    - $||\mathbf{x}||^2 \geq 0$ for any $\mathbf{x}$, so the Hessian matrix is PSD.
    - As a result, the function $f(\mathbf{w}) = \frac{||\mathbf{w}||^2}{2}$ is convex!

$$I_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

Dual Problem

- **Step 2: Make the Lagrange Dual function**
  - We have $\mathbf{w}^* = \sum_{i=1}^{N} \alpha_i y_i \mathbf{x}_i$ and $\sum_{i=1}^{N} \alpha_i\, y_i = 0$
  - Put these things to $L_P(\mathbf{w}, b, \boldsymbol{\alpha})$:

$$L_P(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{||\mathbf{w}||^2}{2} - \sum_{i=1}^{N} \alpha_i \cdot \{y_i(\mathbf{w}^\mathbf{T}\mathbf{x}_i + b) - 1\}$$

$$= \frac{||\mathbf{w}||^2}{2} - \sum_{i=1}^{N} \alpha_i y_i \mathbf{w}^\mathbf{T}\mathbf{x}_i - \sum_{i=1}^{N} \alpha_i y_i b + \sum_{i=1}^{N} \alpha_i \qquad (1)$$

$$\frac{||\mathbf{w}||^2}{2} = \frac{\mathbf{w}^\mathbf{T}\mathbf{w}}{2} = \frac{\mathbf{w}^\mathbf{T}\sum_{i=1}^{N} \alpha_i y_i \mathbf{x}_i}{2} = \frac{\sum_{i=1}^{N} \alpha_i\, y_i \mathbf{w}^\mathbf{T}\mathbf{x}_i}{2} \qquad (2)$$

By integrating (1) and (2), we have

$$L_P(\mathbf{w}, b, \boldsymbol{\alpha}) = -\frac{\sum_{i=1}^{N} \alpha_i\, y_i \mathbf{w}^\mathbf{T}\mathbf{x}_i}{2} + \sum_{i=1}^{N} \alpha_i$$

Dual Problem

- **Step 2: Make the Lagrange Dual function**
  - We have $\mathbf{w}^* = \sum_{i=1}^{N} \alpha_i y_i \mathbf{x}_i$ and $\sum_{i=1}^{N} \alpha_i\, y_i = 0$
  - $\mathbf{w}^{*\mathbf{T}} = \sum_{i=1}^{N} \alpha_i y_i \mathbf{x}_i^{\mathbf{T}}$

$$\sum_{i=1}^{N} \alpha_i - \frac{\sum_{i=1}^{N} \alpha_i\, y_i \mathbf{w}^{\mathbf{T}} \mathbf{x}_i}{2} = \sum_{i=1}^{N} \alpha_i - \frac{\sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^{\mathbf{T}} \mathbf{x}_j}{2} = L_D(\boldsymbol{\alpha})$$

- **Step 3: Find α that maximizes the Lagrange Dual function**

$$\max_{\boldsymbol{\alpha}} L_D(\boldsymbol{\alpha})$$

$$\text{s.t. } \sum_{i=1}^{N} \alpha_i\, y_i = 0 \text{ and } \alpha_i \geq 0, \quad \forall i \in [\mathbf{1}, \boldsymbol{N}]$$

  - This is a kind of QP (quadratic programming). How to solve?
    - Use a well-known solver !

QP Solver

- **We use quadratic programming (QP) to solve the Dual problem**
  - Original QP form:

$$\begin{aligned} \underset{\mathbf{x}}{\text{minimize}} \quad & \frac{1}{2}\mathbf{x}^\top Q\mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ \text{subject to} \quad & A\mathbf{x} \leq \mathbf{b} \\ & E\mathbf{x} = \mathbf{d} \end{aligned}$$

$$\Rightarrow$$

$$\begin{aligned} \underset{\boldsymbol{\alpha}}{\text{minimize}} \quad & \frac{1}{2}\boldsymbol{\alpha}^\top Q\boldsymbol{\alpha} - \mathbf{1}^\top \boldsymbol{\alpha} \\ \text{subject to} \quad & -\boldsymbol{\alpha} \leq \mathbf{0} \\ & \mathbf{y}^\top \boldsymbol{\alpha} = \mathbf{0} \end{aligned}$$

  - SVM's Lagrange dual problem:

$$\begin{aligned} \underset{\boldsymbol{\alpha}}{\text{maximize}} \quad & \sum_{t=1}^{N} \alpha_t - \frac{1}{2}\sum_{s=1}^{N}\sum_{t=1}^{N} \alpha_s \alpha_t y_s y_t \mathbf{x}_s^\top \mathbf{x}_t \\ \text{subject to} \quad & \sum_{t=0}^{N} \alpha_t y_t = 0 \\ & \alpha_t \geq 0, \ \forall t \in [1, N] \end{aligned}$$

QP Solver

- **Conversion of SVM dual form for QP form**
  - The following Q converts SVM dual form into QP form

$$Q = \begin{bmatrix} y_1 y_1 \mathbf{x}_1^\top \mathbf{x}_1 & y_1 y_2 \mathbf{x}_1^\top \mathbf{x}_2 & \cdots & y_1 y_N \mathbf{x}_1^\top \mathbf{x}_N \\ y_2 y_1 \mathbf{x}_2^\top \mathbf{x}_1 & y_2 y_2 \mathbf{x}_2^\top \mathbf{x}_2 & \cdots & y_2 y_N \mathbf{x}_2^\top \mathbf{x}_N \\ \vdots & \vdots & \vdots & \vdots \\ y_N y_1 \mathbf{x}_N^\top \mathbf{x}_1 & y_N y_2 \mathbf{x}_N^\top \mathbf{x}_2 & \cdots & y_N y_N \mathbf{x}_N^\top \mathbf{x}_N \end{bmatrix}$$

  - Then a QP solver will return optimal $\boldsymbol{\alpha} = < a_1, a_2, \ldots, a_N >$

$$\begin{array}{ll} \underset{\boldsymbol{\alpha}}{\text{minimize}} & \frac{1}{2}\boldsymbol{\alpha}^\top Q \boldsymbol{\alpha} - \mathbf{1}^\top \boldsymbol{\alpha} \\ \text{subject to} & -\boldsymbol{\alpha} \leq \mathbf{0} \\ & \mathbf{y}^\top \boldsymbol{\alpha} = \mathbf{0} \end{array}$$

Final Step

- **Final solutions to obtain our decision boundary** $g(\mathbf{x}) = \mathbf{w}^{\mathsf{T}}\mathbf{x} + b$
  - $$\boxed{\mathbf{w} = \sum_{i=1}^{N} \alpha_i \, y_i \mathbf{x}_i}$$
  - $$\boxed{\mathbf{w}^{\mathsf{T}}\mathbf{x}_+ + b = 1 \quad \& \quad \mathbf{w}^{\mathsf{T}}\mathbf{x}_- + b = -1}$$

    $$=> \; b = -\frac{1}{2}(\mathbf{w}^{\mathsf{T}}\mathbf{x}_+ + \mathbf{w}^{\mathsf{T}}\mathbf{x}_-)$$

- **Remember the "complementary slackness" condition!**
  - $$\boxed{\alpha_{\mathrm{i}}\left(y_{\mathrm{i}}\left(\mathbf{w}^{\mathsf{T}}\mathbf{x} + b\right) - 1\right) = 0}$$

Optimization

- **From the complementary slackness condition, we know that:**

$$\alpha_i\left(y_i\left(\mathbf{w}^{\mathrm{T}}\mathbf{x}_i + b\right) - 1\right) = 0$$

  - Non-support vectors: $\alpha_i = 0$
  - Only the **support vectors** have $\alpha_i \neq 0$
  - Then, the solution has the form:

$$\mathbf{w} = \sum_{i=1}^{N} \alpha_i\, y_i \mathbf{x}_i = \sum_{\boxed{i \in SV}} \alpha_i\, y_i \mathbf{x}_i$$

- **Classification**
  - For a new data $(\mathbf{x}_{new}, y_{new})$, the prediction can be done by:

$$y_{new} = sign(\mathbf{w}^{\mathbf{T}}\mathbf{x}_{new} + b)$$

Next Topics

- **Hard Margin SVM**
  - It works only for the ideal case (linearly separable data)

- **How to make it work for linearly non-separable data?**
  - Option 1: Allow the error cases and penalty them

  => Soft margin SVM!

  - Option 2: Non-linear decision boundary

  => Kernel trick!

Thank You